

Computational Extension of OEIS Sequence A163665: Eight New Terms via a Segmented Prime-Index Search

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Abstract

OEIS A163665 consists of the integers k such that

$$k = \text{prime}(\varphi(\sigma(\pi(k)))).$$

Since every term is prime, the search can be transferred to the prime index $n = \pi(k)$: an index n is a hit precisely when $\varphi(\sigma(n)) = n$, and the corresponding sequence term is $\text{prime}(n)$. This note reports a segmented ISO/IEC 9899:2018 C (C17) [7] computation over

$$234954224 \leq n \leq 405234954223$$

the range following the earlier OEIS bound $k \leq 5 \cdot 10^9$. The campaign found eight new terms and no further terms through

$$\text{prime}(405234954223) = 11776117514717$$

The negative certification is finite and computational: it holds subject to the correctness of the segmented production program and the integrity of the recorded run manifests. All 28 positive rows in the consolidated b-file were also checked independently with PARI/GP for $\varphi(\sigma(n)) = n$ and primality of k , and with `primecount` for the identity $k = \text{prime}(n)$; this includes the eight new terms.

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1 Introduction

Let $\text{prime}(m)$ denote the m -th prime, let $\pi(x)$ denote the prime-counting function, let φ be Euler's totient function, and let σ be the sum-of-divisors function. OEIS A163665 is defined by

$$k = \text{prime}(\varphi(\sigma(\pi(k))))$$

The OEIS snapshot used for this computation lists the first 20 terms, ending at

$$a(20) = 2444540149$$

and records that there is no further term up to $5 \cdot 10^9$ [1]. Since

$$\pi(5 \cdot 10^9) = 234954223$$

the next natural computational task is to search prime indices $n > 234954223$.

The main contribution of this note is a finite computational extension. The production campaign scans the contiguous interval

$$234954224 \leq n \leq 405234954223$$

and reports exactly eight hits. Translating those hits back by $k = \text{prime}(n)$ gives

$$\begin{aligned} &50685770167, \quad 94206075193, \quad 106882008719, \quad 571264143487, \\ &1272242879459, \quad 2536961181761, \quad 3715374607207, \quad 9576704442589. \end{aligned}$$

The computation also certifies that there is no further term after 9576704442589 up to

$$11776117514717$$

This is a computational note rather than a theoretical classification. No density law, infinite family, or proof of eventual behavior is claimed. The finite exclusion interval is certified by the C17 segmented kernel, the recorded run manifest, and the validation artifacts described below.

2 Definition and reduction to prime indices

Definition 1. An integer $k \geq 2$ belongs to A163665 if

$$k = \text{prime}(\varphi(\sigma(\pi(k))))$$

Proposition 2 (Prime-index reduction). *Let $p_n = \text{prime}(n)$. An integer k belongs to A163665 if and only if $k = p_n$ for some $n \geq 1$ satisfying*

$$\varphi(\sigma(n)) = n$$

Proof. If $k \in \text{A163665}$, then the right hand side of the defining equation is a prime, so k is prime. Hence $k = p_n$, where $n = \pi(k)$. The defining equation becomes

$$p_n = \text{prime}(\varphi(\sigma(n)))$$

and injectivity of the prime-index map gives $\varphi(\sigma(n)) = n$. Conversely, if $\varphi(\sigma(n)) = n$ and $k = p_n$, then $\pi(k) = n$ and therefore

$$\text{prime}(\varphi(\sigma(\pi(k)))) = \text{prime}(\varphi(\sigma(n))) = \text{prime}(n) = k$$

□

Lemma 3 (Parity filter). *Every solution $n > 1$ of $\varphi(\sigma(n)) = n$ is even.*

Proof. For $n > 1$, one has $\sigma(n) > 2$. Euler's totient is even for every integer greater than 2. Thus $\varphi(\sigma(n))$ is even, and so n is even whenever $\varphi(\sigma(n)) = n$. □

Lemma 3 is the only global pruning rule used in the certified scan. Stronger congruence restrictions were not assumed.

3 Computational method

The production program is the single C17 source file `src/a163665.c`, built by `src/Makefile`. It provides three modes:

```
selftest,      segment -lo N -hi M -threads T -out PREFIX,      summarize.
```

The search mode treats the range $[N, M]$ as inclusive. Within each segment it retains $n = 1$ and even $n > 1$, in accordance with Lemma 3.

For each retained candidate n , the kernel performs the following exact computation.

1. Factor n within the segment and compute $\sigma(n)$ from its factorization. The source builds a prime table up to $\sqrt{hi}$, visits the multiples of each prime in the inclusive segment, removes prime powers from a mutable residual array, and multiplies the corresponding divisor-sum factors into a checked 64-bit accumulator. Any residual factor larger than 1 is then accounted for as a prime factor of n .
2. Factor $\sigma(n)$ using the 64-bit factorization routines in the production source. These first remove prime factors up to 97, then use a deterministic Miller–Rabin test with bases

2, 325, 9375, 28178, 450775, 9780504, 1795265022

and a fixed-seed Pollard–Rho recursion for composite cofactors [3, 4, 5, 6]. This base set is recorded as deterministic for at least the unsigned 64-bit range; the implementation reduces each base modulo the candidate before testing. The Pollard–Rho routine tries constants $c = 1, \dots, 32$, uses the iteration $x \mapsto x^2 + c \pmod{n}$ with Floyd cycle detection, and declares a candidate unresolved if factorization fails. Arithmetic is checked in unsigned 64-bit storage, with 128-bit intermediates for products and modular arithmetic; an overflow or factorization failure is therefore a segment failure or unresolved candidate, not a skipped case.

3. For every prime divisor q of $\sigma(n)$, reject the candidate if $q - 1 \nmid n$. This is a necessary condition, not a heuristic: if $\varphi(\sigma(n)) = n$, then each factor $q - 1$ occurring in $\varphi(\sigma(n))$ divides n .
4. Compute $\varphi(\sigma(n))$ exactly from the factorization of $\sigma(n)$. If the result equals n , record a hit.
5. Convert a hit index to a sequence term by computing `prime(n)`. The production source uses an internal Lehmer prime-counting routine for small hit indices and delegates larger hit-index conversion to the command-line program `primecount`; the public validation script independently rechecks every reported mapping with `primecount -nth-prime -threads=1`.

Each segment writes a hit table and a manifest. The manifest records the range, parity policy, tested-candidate count, rejected-candidate count, factorization-path telemetry, hit count, status, elapsed time, and file checksums. The source and output artifacts are protected by SHA-256 fields; compact FNV-1a 64-bit fields are also recorded as secondary checksums. Only segments with status `OK_EMPTY` or `OK_HITS`, zero unresolved candidates, and zero failed candidates are used in the certified negative result. The manuscript does not claim formal verification of the C implementation or a second exhaustive scan of the non-hit interval.

4 Validation and independent checks

The public package records the final validation state in `validation/validation_summary.md`. The final campaign synthesis uses official runs `A163665_run_001` through `A163665_run_015`; the pilot overlap run `A163665_run_000` and interrupted old-binary artifacts are excluded.

The final validation report gives the following aggregate counts:

official runs	15
certified subsegments	40500
raw n -width	405000000000
candidates tested	202500000000
unresolved segments	0
failed segments	0
new hits	8

The elapsed total computed from the official run logs is `22h:12m:58s`. This is the elapsed total reconstructed from the official run-log time fields, not a mathematical input and not a portable benchmark. The public aggregate manifest records run-level coverage, hit counts, and failure/unresolved status; it does not by itself preserve every per-subsegment timing and thread-count field.

The local validation environment for this package was Linux Mint 22.3 on an AMD Ryzen 9 7940HS system with 8 cores, 16 hardware threads, and 64 GB DDR5 installed RAM. The public Makefile builds with GCC 13.3.0 using

```
-std=c17 -O2 -Wall -Wextra -Wshadow -Wformat=2
```

together with the additional warning flags in `src/Makefile` and `-fopenmp`. The independent scripts were run with PARI/GP 2.15.4 and primecount 7.10 in the local validation environment. These version strings document the package validation environment; the public archive for this package is release `v1.0.1`.

The validation matrix is:

Check	Recorded result
C selftest: b-file terms, known index witnesses, low-range cross-check, and segmented/direct comparison	selftest OK
Global run-summary validation	bad =0, contiguous coverage
Official submanifest count	40500 files
One-point reproduction of the largest discovered hit	OK_HITS at $n = 331914240000$
PARI/GP positive-hit verification	all 28 terms OK
primecount prime-index mapping verification	all 28 mappings OK

The PARI/GP script `validation/verify_hits.gp` recomputes $\varphi(\sigma(n)) = n$ for all 28 consolidated positive indices and checks primality of the corresponding k -values. Thus the independent positive checker covers the 20 previously listed OEIS terms as well as the eight new discoveries. The shell script `validation/verify_prime_mapping.sh` uses primecount to verify $k = \text{prime}(n)$ for the same 28 rows. These scripts are independent positive-hit checks. They do not constitute a second exhaustive scan of the non-hit interval. The negative certification is therefore bounded by

the correctness of the production kernel and the integrity of the recorded segment manifests. The endpoint value used for the final bound,

$$\text{prime}(405234954223) = 11776117514717$$

is recorded in `validation/validation_summary.md` and is reproduced by

```
primecount 405234954223 -nth-prime -threads=1
```

The PARI/GP checker intentionally avoids $\pi(k)$ or $\text{prime}(n)$ calls at the largest indices, because those are not lightweight checks in PARI/GP; the prime-index mapping is checked separately with `primecount`.

5 Results

Theorem 4 (Validated production search result). *Subject to the correct execution of the validated C17 production kernel and to the integrity of the recorded official run manifests, the only solutions of*

$$\varphi(\sigma(n)) = n$$

in the interval

$$234954224 \leq n \leq 405234954223$$

are

$$\begin{aligned} &2147483648, \quad 3889036800, \quad 4389396480, \quad 21946982400, \\ &47416320000, \quad 92177326080, \quad 133145026560, \quad 331914240000. \end{aligned}$$

Their corresponding A163665 terms are the eight values listed in Table 1.

Proof, computational. The interval was partitioned into the 15 official runs listed in Appendix A. The run-level manifest `validation/run_manifest.tsv` records contiguous coverage from 234954224 through 405234954223, zero unresolved segments, zero failed segments, and eight total hits. The final validation report records 40500 certified subsegments and confirms the same global coverage and hit count. The hit rows in `results/A163665_campaign_hits.tsv` and `data/certified_terms.tsv` are identical up to comment/header lines. The independent PARI/GP and `primecount` scripts verify all 28 positive rows in the consolidated b-file, including the eight new rows reported in Table 1. $\square$

$a(i)$	n	$k = \text{prime}(n)$	$\sigma(n)$
21	2147483648	50685770167	4294967295
22	3889036800	94206075193	16331433888
23	4389396480	106882008719	18377794080
24	21946982400	571264143487	94951936080
25	47416320000	1272242879459	204721968000
26	92177326080	2536961181761	386940247200
27	133145026560	3715374607207	601662398400
28	331914240000	9576704442589	1433565580920

Table 1: The eight new certified terms of OEIS A163665. For each row, the certified equality is $\varphi(\sigma(n)) = n$.

Corollary 5 (Extended OEIS bound). *With the same computational assumptions as in Theorem 4, and using the previously recorded OEIS verification that no further term occurs for $k \leq 5 \cdot 10^9$, the terms of A163665 not exceeding 11776117514717 are exactly the 28 terms in the consolidated b-file, ending at*

$$a(28) = 9576704442589$$

and there is no further term in

$$9576704442589 < k \leq 11776117514717$$

Proof. By Proposition 2, terms k of A163665 in the searched range are exactly $k = \text{prime}(n)$ for the index hits n in Theorem 4. The final endpoint is recorded as

$$\text{prime}(405234954223) = 11776117514717$$

Since the previous OEIS-style bound corresponds to $\pi(5 \cdot 10^9) = 234954223$, the certified index interval begins immediately after that bound. $\square$

The file `data/b163665.txt` is therefore the candidate extended OEIS b-file associated with this computation.

6 Observed structure and conjectural remarks

The eight new hit indices have relatively smooth factorizations; see Appendix B. Several rows contain a common factor 2^{13} , and the factorizations of $\sigma(n)$ repeatedly involve small primes such as 43 and 127. In these eight new rows, the largest prime factor of n is at most 31. The first new hit,

$$n = 2147483648 = 2^{31}$$

is especially transparent:

$$\sigma(n) = 2^{32} - 1 = 3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$$

and the product of $q - 1$ over the distinct prime factors q gives $\varphi(\sigma(n)) = 2^{31}$. Thus the largest prime factor of $\sigma(n)$ over all eight new rows is 65537, while among the remaining seven rows it is 2801.

These features are reported only as observations. They were not used as filters in the certified search, and they do not presently constitute an infinite-family construction or a density model. A cautious conjectural reading is that future hits, if any, may continue to favor indices whose factorization makes $\sigma(n)$ highly structured. No claim stronger than this is supported by the present computation.

7 Reproducibility

The public release package contains the production source, Makefile, data tables, validation reports, and independent scripts required to audit the finite computation. The main local artifacts are:

<code>src/a163665.c</code>	C17 production source,
<code>src/Makefile</code>	build, test, and sanitizer targets,
<code>docs/confirmation_A163665.md</code>	OEIS snapshot excerpt and triage confirmation,
<code>docs/strategy_A163665.md</code>	computational strategy notes,
<code>validation/run_manifest.tsv</code>	official run-level manifest,
<code>validation/validation_summary.md</code>	final validation report,
<code>data/b163665.txt</code>	candidate extended OEIS b-file,
<code>data/certified_terms.tsv</code>	new terms with witnesses,
<code>results/A163665_campaign_hits.tsv</code>	consolidated hit table,
<code>validation/verify_hits.gp</code>	PARI/GP positive-hit checker,
<code>validation/verify_prime_mapping.sh</code>	primecount mapping checker.

The package validation report records the following commands as passed:

```
make -C src clean all
make -C src test THREADS=4
make -C src sanitize
gp -q validation/verify_hits.gp
validation/verify_prime_mapping.sh
make -C src clean.
```

The large local subsegment directories generated during the campaign are not included in the public package. Their aggregate certified content is represented by the run manifest and validation summary. Release v1.0.1 is archived with Zenodo DOI <https://doi.org/10.5281/zenodo.21242104>.

8 Conclusion

The computation extends OEIS A163665 from 20 to 28 known terms. The new terms are the eight primes listed in Table 1, and the certified absence of additional terms now reaches

$$k = 11776117514717$$

The mathematical reduction to the fixed-point condition $\varphi(\sigma(n)) = n$ is exact, while the extension itself is a finite computational result. The main limitation is also explicit: the positive rows have independent PARI/GP and primecount checks, but the exclusion of non-hits over the full interval rests on the validated segmented C17 campaign and its recorded manifests.

Data availability

The code, OEIS snapshot excerpt and confirmation notes, result tables, manifests, validation summaries, candidate b-file, and independent positive-hit checkers are part of the public project package archived in the GitHub repository and Zenodo record listed in the References. The snapshot material is recorded in `docs/confirmation_A163665.md`, not as a separate standalone input-snapshot file. No additional dataset is needed to reproduce the finite claims stated in this note, apart from standard toolchain components and the external programs used by the recorded validation commands.

AI use disclosure

Large language models were used as research and drafting assistants during the A163665 workflow, including OpenAI Codex and external LLM-based review systems from the ChatGPT, Claude, DeepSeek, Gemini, Grok, Kimi, Qwen, MiniMax and Z families. NotebookLM was also used as a research-assistant tool during the methodological workflow. These systems are named at family or tool level where exact model-version metadata were not consistently available in the local review artifacts.

Project workflow. LLM assistance was used to organize the OEIS snapshot, triage the equivalence between the original definition and the prime-index fixed-point problem, compare possible computational strategies, and prepare publication-facing notes from the consolidated package. Internal triage judgments and convenience assessments are not used as mathematical evidence in this paper.

Mathematical analysis. LLM assistance was used to structure the reduction from k to $n = \pi(k)$, the parity filter, the exact $(q - 1) \mid n$ rejection condition, and the distinction between theorem, computation, and conjectural observation. The mathematical acceptance criterion is the explicit equality $\varphi(\sigma(n)) = n$, checked against the local OEIS snapshot, the result tables, and the independent PARI/GP positive-hit verifier.

Algorithm and code. LLM assistance was used in the design, explanation, and review loop for the C17 segmented implementation. The final numerical claims rely on the local source, Makefile, run manifest, validation summary, b-file, known-term replay, sanitizer run, one-point reproduction of the largest new hit, and independent positive-hit scripts, not on conversational assertions.

Manuscript. LLM assistance was used to draft and structure this manuscript from the project artifacts. Numerical claims in the manuscript were checked against the local b-file, certified-term table, campaign hit table, run manifest, validation notes, validation summary, and independent verifier scripts.

The author takes full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries, and conclusions of this paper.

A Official run manifest summary

Run	n -interval	Certified subsegments	Hits
run_001	234954224–30234954223	3000	4
run_002	30234954224–60234954223	3000	1
run_003	60234954224–90234954223	3000	0
run_004	90234954224–120234954223	3000	1
run_005	120234954224–150234954223	3000	1
run_006	150234954224–180234954223	3000	0
run_007	180234954224–205234954223	2500	0
run_008	205234954224–230234954223	2500	0
run_009	230234954224–255234954223	2500	0
run_010	255234954224–280234954223	2500	0
run_011	280234954224–305234954223	2500	0
run_012	305234954224–330234954223	2500	0
run_013	330234954224–355234954223	2500	1
run_014	355234954224–380234954223	2500	0
run_015	380234954224–405234954223	2500	0

Table 2: Official run-level manifest summary.

Every official run in the public manifest has `coverage=CONTIGUOUS`, `segments_unresolved=0`, and `segments_failed=0`.

B Factorization witnesses for the new hits

n	Factorization of n	Factorization of $\sigma(n)$
2147483648	2^{31}	$3 \cdot 5 \cdot 17 \cdot 257 \cdot 65537$
3889036800	$2^9 \cdot 3^4 \cdot 5^2 \cdot 11^2 \cdot 31$	$2^5 \cdot 3 \cdot 7 \cdot 11^3 \cdot 19 \cdot 31^2$
4389396480	$2^{13} \cdot 3^7 \cdot 5 \cdot 7^2$	$2^5 \cdot 3^3 \cdot 5 \cdot 19 \cdot 41 \cdot 43 \cdot 127$
21946982400	$2^{13} \cdot 3^7 \cdot 5^2 \cdot 7^2$	$2^4 \cdot 3^2 \cdot 5 \cdot 19 \cdot 31 \cdot 41 \cdot 43 \cdot 127$
47416320000	$2^{13} \cdot 3^3 \cdot 5^4 \cdot 7^3$	$2^7 \cdot 3 \cdot 5^3 \cdot 11 \cdot 43 \cdot 71 \cdot 127$
92177326080	$2^{13} \cdot 3^8 \cdot 5 \cdot 7^3$	$2^5 \cdot 3^2 \cdot 5^2 \cdot 13 \cdot 43 \cdot 127 \cdot 757$
133145026560	$2^{13} \cdot 3^6 \cdot 5 \cdot 7^3 \cdot 13$	$2^6 \cdot 3^2 \cdot 5^2 \cdot 7 \cdot 43 \cdot 127 \cdot 1093$
331914240000	$2^{13} \cdot 3^3 \cdot 5^4 \cdot 7^4$	$2^3 \cdot 3 \cdot 5 \cdot 11 \cdot 43 \cdot 71 \cdot 127 \cdot 2801$

Table 3: Factorization witnesses for the eight new hits.

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